HPC-I Chap4-2 Lab1-1 IPM profiler

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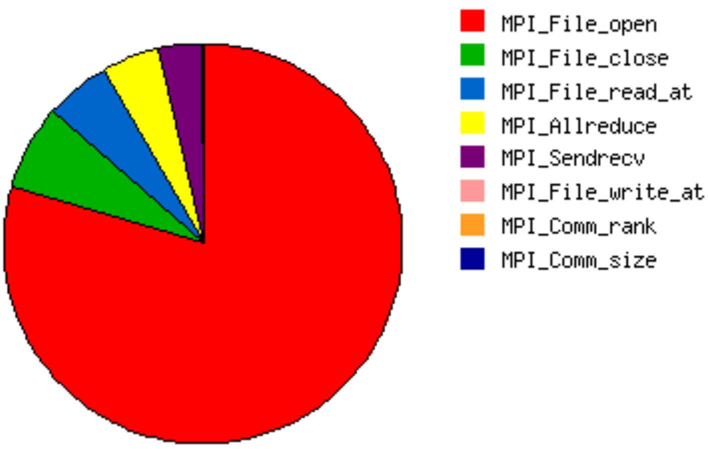
(1) Identify Top MPI Functions:

(a) Examine the profiling data given in profile_me_16. This report was generated using mpirun -np 16 ./profile_me 99999 ./09.in output.bin on an Intel X5670 system (2 sockets × 6 cores). (No need to answer)

(b) List the top three MPI functions that consume the most time in the application. Provide a brief explanation for why you think these functions are the most time-consuming.

Ans:

For profile_me_16, the top three MPI functions are **MPI_File_open**, **MPI_File_close**, and **MPI_File_read_at**. **MPI_File_open** takes the largest portion of the time, followed by **MPI_File_close** and **MPI_File_read_at.** This shows that the program spends most of its time on MPI file I/O, so the main overhead is file access rather than computation or normal communication.



(2) Visualization:

(a) Paste the pie chart generated by the IPM profiler(the one you generated with mpirun -np 2 ./profile_me 99999 ./09.in output.bin) that illustrates the time distribution across different MPI functions in your application.

Ans: